

ALEJANDRO ANGEL

Data Scientist with 4 years of experience analyzing product sales, customer segmentation, and transactional data for industries such as retail/e-commerce and foodtech, healthcare, and banking. Capabilities include time series forecasting using RNNs and LSTMs and owning the entire project lifecycle from data acquisition to informative visualizations. Recent projects include implementing a financial product validation system to verify the ownership of financial products from multiple banks associated with mobile phone numbers, designing and building infrastructure for smooth management & deployment of ML models, and implementing high-performance data pipelines supporting up to 300 transactions/second.

SKILLS

Programming Languages: Matlab, Bash, SQL, Python

Databases & Storage: DynamoDB

Data Tools: Airflow, Pandas, Databricks, PySpark

Visualization Tools: Matplotlib, Seaborn, Plotly

Cloud: AWS

ML Frameworks & Tools: Scikit-learn, spaCy, TensorFlow, Keras

Others: LaTeX, AWS Lambda, FastAPI, AWS Batch, Jupyter, AWS API Gateway, AWS Glue, AWS Step Functions, AWS Quicksight

EXPERIENCE

Data Scientist & Engineer, Masiv/TeleSign

Jan 2022 – Present

- Designed and developed data products for a large omnichannel solution in Latam and India, servicing clients in healthcare, retail, and banking, handling large volumes of data.
- Designed & launched a feature for the company's digital identity product to automatize the third-party data collection process from data coming in monthly.
 - Implemented a data pipeline using AWS Lambdas, AWS Batch Job, ECS, FastAPI, and Polars to collect, process, store, and serve large amounts of data; this resulted in a high-performance service querying multiple sources, supporting up to 300 transactions per second.
- Designed & built infrastructure for smooth model management & deployment, as well as updated and streamlined the original model that powered the digital identity product, eliminating redundant rules.
 - Implemented a LightGBM regressor to estimate anomaly scores of mobile phone numbers based on their transactional history.
 - Utilized AWS API Gateway, AWS Lambda, and ECS for the infrastructure.
- Implemented a financial product validation system to verify the ownership of financial products from multiple banks associated with mobile phone numbers.
 - Utilized spaCy's NER to extract key information from text messages (SMS) for deeper analysis; utilized DynamoDB, AWS Batch Job, AWS API Gateway, and AWS Lambda.

- Updated several custom business logic steps from Pandas to PySpark for a Databricks workflow, optimizing the code integrity and improving execution times by up to 38%.
- Implemented infrastructure as a code with AWS CDK for automating and streamlining cloud resource provisioning.

Data Scientist, Iluma Alliance

Apr 2021 – Jan 2022

- Maintained and developed functionalities for a FastAPI-based product used by the commercial team of a US-based foodtech.
 - Built API endpoints for querying sales data across various product references, enabling efficient data retrieval and analysis.
- Developed a Streamlit dashboard for a commercial representative to track sales performance, analyze product trends, and identify key sales patterns.
 - Implemented customer segmentation and analytics, providing insights into the client base.
- Explored time series forecasting using RNNs and LSTMs with the company sales data, paving the way for future predictive capabilities and actionable insights.
 - Owned the entire project lifecycle, from data acquisition through the e-commerce API to informative visualizations.

Researcher, Universidad Pontificia Bolivariana

Jan 2018 – Jan 2020

- Member of the Numerical Methods and Simulation research group.
- Developed numerical models in Matlab to simulate the operating conditions of construction materials.
- Study of forest fires in the Aburrá Valley processing satellite data with Python.

EDUCATION & TRAINING

PgDip, Data Science, Universidad De Antioquia

Ongoing

BSc, Aeronautical Engineering, Universidad Pontificia Bolivariana

Certifications: **AWS Partner Accreditation (AWS)**

Courses: **Deployment Of Machine Learning Models** (Udemy), **Testing Python Data Science Code** (Linkedin), **Unit Testing And Test Driven Development In Python** (Linkedin), **Introduction To Machine Learning In Production** (Deeplearning), **Introduction To Data Science In Python** (University Of Michigan), **Neural Networks And Deep Learning** (Deeplearning), **Convolutional Neural Networks** (Deeplearning), **Machine Learning** (Stanford University)

PUBLICATIONS

- Suárez, G., Garay, F., Urraya, E., Cruz, L.J., Ortells, L.J., De La Hoz, K., Loaiza P., & Ángel, A. (2020, October). Computational Analysis of the IN718 Material Microstructure (Nickel-Based

INCONEL Superalloy): Potential Applications in the Military Industry. Applied Computer Sciences in Engineering.

- Ángel, A., Suárez, G., & Niño, N. (2020, June). Computational Simulation of the Temperature Distribution of Aerogel-Based Coated and Uncoated Multi-Layer Panels for Thermal Insulation. IOP Conference Series: Materials Science and Engineering.
- Ángel, A., Suárez, G., & Niño, N. (2019, May). Computational Analysis of the Thermal Behavior on a Silica (SiO_2) Aerogel Coating for Applications in the Construction Industry. IOP Conference Series: Materials Science and Engineering.